PII-GUARD MIDDLEWARE

Privacy Layer for AI Agents | Data Privacy | Prototype Submission

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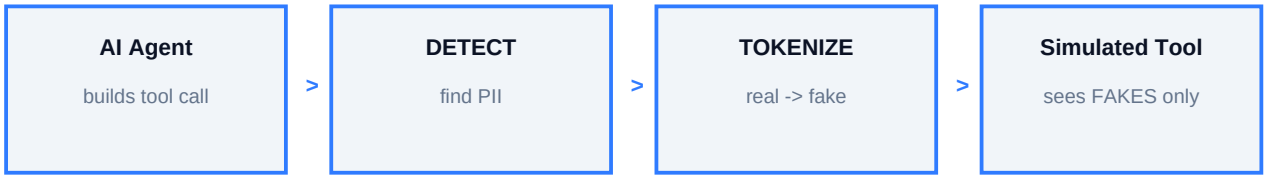
AI agents may use personal information - such as phone numbers, email addresses, and identification numbers - to link user tools through prompts and tool parameters. When an agent calls an external tool (sending an SMS, looking up a record, charging a card), real personal values travel inside those tool requests and get stored in third-party logs. A privacy layer is needed to detect and protect sensitive values before outgoing requests leave the application.

2. Our solution in one line

Invisible checkpoint for AI agents

Every agent-to-tool payload passes through PII-Guard middleware: it detects personal information, substitutes reversible tokens, sends only fakes to the simulated external tool, and restores the correct original values in the response.

3. Architecture



Tool answers flow back through the guard: tokens are restored before reaching the agent.

4. How it protects (4 steps)

1. **INTERCEPT**
   middleware captures the outgoing agent-to-tool payload, both structured JSON and natural-language text.
2. **DETECT**
   finds phone numbers, email addresses, PAN, Aadhaar and debit/credit card numbers via patterns + checksums (Verhoeff for Aadhaar, Luhn for cards).
3. **PROTECT**
   replaces each real value with a format-preserving reversible token; a second scan blocks the request if any real value survives (fail-closed).
4. **RESTORE**
   tool responses containing tokens are mapped back to the correct originals before returning to the agent.

5. What it detects

PII TYPE	HOW DETECTED	FAKE TOKEN EXAMPLE
Email	standard pattern	user_0001@redacted.test
Indian phone	+91 / prefix + 6-9 rule	+91-00000-00001
PAN	ABCDE1234F uppercase	QWERT0001Z
Aadhaar	12-digit + Verhoeff	2999-8888-0001
Debit / credit card	13-19 digits + Luhn	4111 0000 0000 0024

6. Tech stack

- Backend: Python + FastAPI (detectors, token vault, audit log).
- Frontend: React + Vite demo dashboard (website is the demo face; the product is the middleware).
- Tests: pytest, seeded synthetic generator. No external network calls. Synthetic data only.

7. Deliverables checklist

- Working middleware that intercepts tool-bound requests (POST /api/invoke).
- Detection for multiple PII types in structured AND natural-language payloads.
- Redaction + response re-hydration demonstrated live in the dashboard.
- Test results + audit log showing exactly what the simulated tool received.

8. Prototype demonstration

Live demo link:

[PASTE DEMO LINK HERE AFTER DEPLOYMENT]

[ADD IMAGE HERE]

Screenshot 1 - protection verdict (SAFE + detections)

[ADD IMAGE HERE]

Screenshot 2 - what-the-tool-received vs restored panels

[ADD IMAGE HERE]

Screenshot 3 - audit log proving the tool got fakes only

9. Real-world relevance

The same checkpoint pattern protects a bank's SMS agent, a hospital booking bot and a shopping assistant: tools keep working because fakes preserve valid formats (a fake card still passes checksum validation), while vendor logs stay worthless to attackers. The audit log is the evidence that the third party never saw real values.

10. Limits and next steps (honest scope)

- Names/addresses need ML-based detection; current MVP is pattern + checksum rules.
- Production needs an encrypted persistent vault, per-user sessions and key rotation.
- CVV/expiry codes are out of scope: any 3-digit number alone is too ambiguous to detect.
- Next build: enforced agent demo where the agent cannot bypass the guard.

11. Fairness note

Designed, built and tested with synthetic personal data only. No real personal or confidential information was used, submitted, or sent to any external service.

12. Reproduce it

Backend: cd backend > pip install -r requirements.txt > python -m uvicorn app:app --port 8000
Frontend: cd frontend > npm install > npm run dev (open http://localhost:5173)
Tests: cd backend > python -m pytest tests/ -q